Uber Model Project Final Report

In this creation of the model the overall goal was to create a linear regression model that would allow us to predict the price of an uber trip. The Uber data set was selected because of its real-world application I have seen currently at my internship. My internship has allowed me to explore aspects of ride-based data and pricing factors such as traffic conditions, and this is a chance to look at it in a new way. I did research at my internship to gain an understanding of Uber and the way they determine pricing but never understood it on a deeper level. Overall, this felt like it would be a fantastic opportunity to make a project that could be included in my resume and tied into my internship while conveying a sense of curiosity and exploration. The objective is to begin from scratch with our Uber data and create a model following the key steps and practices learned to ensure our model is as accurate as possible. All the while ensuring to observe and deal with any anomalies in areas such as assumptions, outliers, and multicollinearity. Although not perfect, no model is perfect and the overall point is to create a meaningful model that could be used in applications and business models. I hope that Uber pricing behavior can highlight how regression techniques can be applied in practical, data-driven decision-making contexts.

We start our journey of creating an uber model by inspecting our data initially on Excel to gain an understanding of our data for importing and seeing if we need to take any additional steps in this importation of data. Immediately, (see image 1) we notice three key things to take notice of when making our new uber model. Firstly, we notice our Y variable is Quantitative meaning we can continue our linear regression model instead of changing paths to a logistic regression model. Secondly, we have qualitative X variables in Time_of_day, Day_of_Week, Traffic_Conditions and Weather. This causes us to go in deeper to gain an understanding of how many dummy variables will be needed to create for our importing of data. Using filters on the data file allows us to summarize we have 4 qualitative X values that in total have 4, 2, 3, 3 different options. With this information it becomes a matter of picking one of the options per variable to use as an anchor for our dummy variables. I went ahead and chose Morning for Time_of_day, Weekday for Day_of_Week, Low for Traffic_Conditions, and Clear for Weather. Now our 8 dummy variables can be seen when running a simple proc print or frequency command (see image 2) will either be 1 if it's that option and 0 otherwise with all the dummy variables of the same variable having the same 0 for the reference category. This step is vital because converting our variables from qualitative to quantitative allows us to capture the relationship between these variables and Trip_Price. Our regression model can more accurately reflect the relationship in the data to improve its predictions, accuracy and potential.

With importation/dummy creation done the next step can start which involves digging deeper into our data with Data Exploration. Many different methods were used to gain an understanding of the data first by viewing the histograms (see image 3). Noting that most quantitative values had zero skewness apart from Trip_Distance_km which skewed to the right. This signaled that an eye should be kept on this variable. It signified that the data in this variable had most of its values fall within the lower range creating a scenario where the mean > median > mode. This skewness means that it may possibly need to consider transformation of the variable when the time comes. Also looked up was the mean table (see image 4) for more information on the first, second and third quartiles. Doing this allows us to see our data split into ¼ different sets to give an idea of where our data is in terms of variation of possible variations. Like Trip_Distance_km having a first quartile of 13 second quartile of 26 and third quartile of 40. The data we have ranges approximately from low numbers to high numbers. This allows an observation to see how condensed or spread our data is, which as can be seen is not so much. Also looked up was the correlation coefficient to gain a general idea of which variables would be most affected by our X values. As can be seen in graph (see image 5) after looking up the correlation coefficient we see high positive correlation between Trip_Distance_km and trip_price thus this is our high positive correlation due to it having a 0.6. Meaning as price goes up so does Trip_distance_km. Another key thing to note is that there is moderate positive correlation with Per_Km_rate and Trip_Duration_Minutes. The rest had around no correlation, some slightly more positive other slightly more negative. These findings suggest that

distance, duration, and rate based variables are likely to be important predictors in the regression model. Identifying these relationships early could help guide the model in future steps and improve the ability to accurately predict Uber Trip_Prices.

Now we can begin examining our initial full model which includes all our variables. The model will attempt to predict our Y or dependent variables with the association with our X or predictors/independent variables. When inspecting our Full Model (see image 6) we want to focus on three key areas. Our P value, goodness-of-fit and Adj-R-Square. The key thing to note is that at a glance we see already see nine variables that are above 0.05 making us see that they are insignificant. Those five being Passenger_Count followed by tod_afternoon, tod_night, dow_weekend, traffic_medium, tod_evening, base_fare, weather_rain and weather_snow. This is important to note for our model because currently these variables do not show a statistically significant relationship to our Y variable and predicting it. If we were to make our model based off this, we would simply remove the largest P value which in this case is Passenger_Count and recheck our P value until all variables are below 0.05. The next area to focus our attention on is the Adj R-square being 0.4945 and R Square being 0.5032. Adj-R squared does not increase with additional X-variable that does not improve the model. These two statistics converted into percentage tell us that our full model with all our variables can explain around 49.45% of our present variability in the Uber data in relation to Trip_Price. Moving onto R^2 is the same explanation except it changes regardless of the variable inserted being significant or not so in this case 50.32% of the variability in the data can be explained using all our variables even those considered insignificant. This is especially important for our model because it allows us to see the scope on which our model can explain the data presented in the Uber file. When applying a model to a business it is important to have a model that can explain the majority of the data so that it can offer a better prediction. Finally, we can take notice of our goodness-of-fit test or GOF. This is an examination to ensure we have at least one variable that is significant. Since our P value is <.0001 we can assume that there is at least one x-variable that has a significant effect on Y and we have a decently high F value of 57.60. With a low P value and an F value of 57.60 we can conclude that the model is highly statistically significant overall meaning we can reject our null hypothesis. This is important because it ensures going forward when we are making our final model for prediction, we will have at least one variable that can predict our Y variable in a significant way.

Now that we have an understanding of our full model we can move onto checking the full models diagnostics. These diagnostics are truly important and will be checked at both the full model and our final model. Now onto our first check in diagnostics being Multicollinearity which is when two or more predictors or X values are highly correlated(related) with each other. This can cause serious issues in our model because it could lead to the coefficient estimates becoming unstable as more predictors are added. Overall, this makes interpreting the regression coefficient difficult because adding a new variable would give us completely different results when it should not be in some areas. This could happen for many reasons such as mis made Dummy variables, insufficient data, or one x value is genuinely made from another x value. Multicollinearity can be checked using the VIF and Tolerance values (see image 7) present in all our variables. Observing the graph allows us to see all the VIF and Tolerance present in the variables. We can see that our VIF is below 10 and tolerance is > 0.10, all remaining high at .9-.6. This means we can remain assured that multicollinearity is not present. Which can reassure us that the result we produce will not be excessively correlated and that each variable contributes to the model. Alongside this the regression coefficients can be interpreted more reliably, and the model is less likely to produce unstable or misleading estimates. Very key to ensuring our model is accurate and reliable in a business practice, reliability is key and we wouldn't want our model falling apart simply because a new variable is added.

Next, we will focus on our outliers and influential points present in our data. Removing these points can significantly improve our model statistical measurements. After scanning through the Studentized Residuals and High cook's D distance and taking note of all the points where it appears to be

higher than expected. They were removed in baches (see image 8) until it could be seen to have stopped causing a significant improvement in Adj R^2. This is important because removing influential points and outliers here affects our models generalization concerns. It won't try and fit the model for all possible points just the average ensuring accuracy in that area. This ensures that later on when creating our prediction it won't affect our model accuracy. As shown in the results (see image 8), the adjusted R-square increased from 0.4945 to 0.5761, then to 0.7611, 0.8542, and finally 0.8864. This show of improvement shows that removing influential points allowed the model to better capture the relationship between the predictors and trip price. However, once the improvement began to stabilize, further removal was no longer necessary, preventing excessive data reduction due to no significant improvement in model. This ensures that the model remains both accurate and representative of the overall dataset without the influence of oddities in our data.

Finally, we will check our 4 assumptions of Constant Variance, Independence, Linearity and Normality. Firstly, the examination will begin by focusing on two key types of graphs that consist of a studentized residuals vs x value and studentized residuals vs predicted values. Let's start with the two assumptions Constant Variance and Independence because we can find our answers on the same graphs (see image 9 and 10) those being our studentized residuals vs predicted values and Studentized residuals vs each x-variable. We will look at the graphs and ensure that they remain random with constant variance, ensuring that there is no spread that shows an increasing or decreasing pattern as we move more to the right on the graph. We notice immediately a pattern on Studentized Residual vs Trip_Distance_km a pattern forming a shape of a curved time pattern. Alongside this discovery on Residual vs Predicted value we see the forming of a clear U shape. These are clear violations that must be addressed in transformation. For constant variance/independence a violation left alone could result in incorrect confidence intervals or prediction intervals, the backbone of our models' predictions confidence. This could lead us into a situation where a business trying to predict the uber cost to set aside money won't have an accurate range to save. Onto Linearity we will be looking at our scatterplot for each x variable vs y variable (see image 11) paying close attention to see if there is a pattern of spread that show a straight line. We see a clear straight line forming as we move to the right of the graph we see an increase along our x value. If we did have a violation in linearity it could result in misleading conclusions so important to ensure its fine. Finally at Normality we can look at the normal probability plot of the residuals (see image 12). Paying attention to ensure that we see something close to a straight line and no curves. Upon inspection everything appears to be okay with our normality having a clear straight line. If we did have an issue in normality, we could have similar issues to constant variance/independence. So, to summarize we passed Linearity and normality but failed constant variance and independence. These issues found in our assumption could put our whole model into doubt and cause surprises down the line so it's important to focus on them and ensure we fix them.

To tackle the violations in our assumptions observed in our initial preview of the full model, transformations were applied to both the X and Y variables. The overall goal was to improve key regression assumptions, particularly those violating independence and Constant variance. Before our transformation the studentized residual plots vs predicted showed a U-shaped pattern or something similar to a curved time pattern can be seen in the studentized residuals vs Trip_Distance_km. This leads to a strong suggestion that the model was not properly capturing the relationship between the X and Y variables. After applying square-root and logarithmic transformations to both X and Y, the studentized residual plots showed significant improvement (see image 13 and 14). The residual plots became more randomized and distributed around the zero line with no pattern, indicating that constant variance and independence have been satisfied. Additionally, we also see that the relationship between the Y variable vs X variable (see image 15) remains the same forming a line. Unfortunately, this improvement in 3 of our assumptions comes at the cost of our normal probability plot(see image 16) showing a slight curvature after transformation. However, I went ahead and accepted this as a tradeoff because we managed to secure Constant variance and Independence which seemed to have huge issues in shapes. A

slight change in curvature seems fine compared to other scenarios of transformation resulting in a blatant S shape in our normal probability. Overall, the transformations allowed the model to improve our model reliability and more accurate predictions of Uber trip prices. With these issues addressed we can feel more confident moving forward that the model will be somewhat more accurate with its predictions and intervals of confidence.

After thoroughly reviewing the checks made on the full model, The process to achieve our final model can begin which we will use to make our predictions. The first step involves splitting the data into two different data sets of 75% training and 25% test. The training set will be used to estimate the best combination for our final model. The test set we create will be used to test predictive performance to ensure our model is a good model. With our new training set of data, we can move forward with our model selection using the training set of data. Deciding to go ahead and choose to use Forward selection (see image 17 left graph) and Backward selection (see image 17 right graph) which gave me very similar results in some areas and slight differences in others. In Forward Selection (see image 17 left graph) the focus is on 3 key statistics R-Square which is 0.8798, F value which is 433.21, and P value for significance which in this case we see 4 variables that are above 0.05 and are insignificant. This model conveys that it has insignificant variables that can explain around 87.97% of our Uber data. For our backward Selection (see image 17 right graph) we see an R-Square of 0.8787, F value of 778.33, P value for all variables is well below 0.05. Similarly, this conveys that we only have significant variables that can explain 87.87% of the Uber data. Comparing both of our models made by different selection methods it becomes apparent that forward selection is better in terms of R^2, F value is better for Backward, and for our p value only backward contains significant variables. They also diverge when it comes to the number of variables Forward selection contains 9 variables with some being insignificant with their p value being around 0.4 or 0.3. Backward selection on the other hand contains 5 significant variables, all of which are very significant, with the highest being 0.0232. The decision is clear and chose the Backward selection model because it removes unnecessary variables that do not contribute meaningfully to predicting Uber trip_price. This results in a simpler and more interpretable model without sacrificing predictive performance.

Now with the final model (see image 18) chosen the process to repeat what was done for examining the full model can restart. Starting with our P value after looking at the output we can see all 5 variables are highly significant with most being at a p value of <.0001 apart from one being 0.02. Compared to the full model where it can be seen to have multiple variables being insignificant this is a huge improvement. There is also a huge reduction in the number of variables, increasing the interpretability of the model by narrowing the number of variables present. The next area to pay close attention to is the Adj R^2 being 0.8226 and R Square being 0.8242. This is a huge improvement from the full model which had an Adj R^2 0.4945 hugely increasing the explanation power of the model in being able to predict our data. This translates to our final going from 49.45% to 82.26% in being able to explain the data present in our Uber data. Finally, the attention can shift to taking notice of the goodness-of-fit test. Since the P value is <.0001 this indicates can be made that there is at least one x variable that has a significant effect on Y which can be seen to be the case. With a low P value and an F value of 531.60 it can be concluded that the model is highly statistically significant overall meaning the null hypothesis can be rejected. This is also another area where a significant increase can be seen going from 57 to 531 in our F value showcasing an even more statistically significant model. Another area to focus attention on is the root mean square where we have 0.76042 compared to the full models 28.816. This is significant because it shows the amount of overall error deviation in our model which is highly significant. This is because it means when for example predicting Trip_Price we won't be off by $800 which is hugely beneficial when trying to predict our uber prices. No one wants to be told that an uber will cost $800 when it is $18 it makes them not want to use uber and is meaningless for both customers and the company. The point in being close to real value is to ensure that someone using this model can make an assumption or decision knowing that it is made based off information that won't be far off.

Now onto the first diagnostic check being to ensure no Multicollinearity is present in the model. This can again be checked by checking the VIF and Tolerance values (see image 19) of the variables in the final model. For these statistics found in the graph (see image 19) we can see that our VIF is below 10, which can be seen in all the variables. All the variables present in the graph are around 1 in terms of variance inflation. When it comes to examining tolerance it can be seen that all variables are above 0.10 with them all remaining high at 0.9 tolerance. This means we can remain assured that multicollinearity is not present. This again helps reassure that the result produced will not be excessively correlated and that each variable contributes to the model. Alongside this the regression coefficients can be interpreted more reliably, and the model is less likely to produce unstable or misleading estimates.

Now onto the second diagnostic check being on the outliers and influential points present in the Uber data of our final model. After a thorough scan through the Studentized Residuals and High cook's D distance and taking note of all the points where it appears a High Cook >1 and a Studentized residuals outside the (-3, 3) scope it's noted that both outliers and influential points are present. However, as can be seen (see image 20 top graph is first batch) after removing the second batch (see image 20 bottom is second) and seeing an increase in Adj R^2 of 0.03 and it becomes a matter of deciding if that is significant enough to keep removing variables. Analyzing the final models original Adj R^2 it can be decided that it isn't worth removing any more data because it will not cause a significant increase in the overall model. The overall Adj R^2 of .9301 from .9040 of the first batch a mere .03 increase wasn't worth sacrificing more data points at the fear of giving my model less valuable information to form an accurate prediction. It also leaves an assurance that the Uber model's accuracy and regression coefficients at this point should no longer be significantly affecting the model by any remaining outliers or influential points.

Finally, we will check our 4 assumptions of Constant Variance, Independence, Linearity and Normality. We'll start again with both Constant Variance and Independence because we can find our answers on the same graphs those being our Studentized Plot residuals vs predicted values and Studentized Plot residuals vs each x-variable. We immediately notice no pattern on both graphs such as Residual vs Trip_Distance_km (see image 21). Also, on Studentized Residual vs Predicted value (see image 22) is showing a random pattern along the 0 with no clear pattern visible. Linearity can be seen looking at the scatterplot for each x-variable (see image 23). Paying close attention to see if there is a pattern of spread that show a straight line. It can be seen in the picture a clear straight line forming as we move to the right of the graph we see an increase along our x value. Finally at Normality we can look at the normal probability plot of the residuals(see image 24). Paying attention to ensure that we see something close to a straight line, and we see a somewhat curve near the middle less prominent than the full model but still present. This is acceptable because we have constant variance, linearity and independence assumptions passing. Overall, the final model meets the necessary four regression assumptions supporting its suitability for making accurate predictions.

Now that we've reviewed our final model we can move onto our model validation and creating our prediction. We use our model to create two separate tests using both our training data and test data comparing the two to ensure we can evaluate the predictive performance of the regression model. We can do this by comparing the computed predictions with the observed values. It is important to note that since we did sqrt (Trip_Price) in transformation all the values will be presented in that fashion, for simplicity of the picture I will keep the values but can be transformed to normal using y^2. Let's start with Test data set (see image 25 graph on left) and computing understanding what are the key numbers we should be looking for this comparison. First RMSE as can be seen here in the picture is 0.82227 meaning that our prediction will generally be off by $0.8 on average. Moving onto MAE, the average difference between predicted and actual values here being $0.51357. Meaning on average we will be off by around $0.5. This is key to understanding how off our prediction will be. For our uber model we want to ensure we are as close as possible to being correct and not off with hundreds of dollars. R^2 being computed by taking the correlation coefficient of 0.89965 and squaring it to get us 0.8092. Now onto our adj R^2 after using the formula we can see we get an answer of 0.8049. This basically explains that our current model can

explain 80.49% of the data found in the test data set. In the case of adj R^2 the higher the better because it explains the variation in the values of Y that is explained by our model. A key feature that models would want our uber model is being used to predict prices possibly for the app the customers use. The more scenarios our model can explain the better it will be at predicting the correct price. Finally, the last number we need for this comparison is CV-R^2 which we get from subtracting our R^2 from training set to R^2 of testing which we will see in a bit will result in 0.1215. This is used to ensure our model is not overfitting the training data and can be generalized to explain the new data. So, in this case since ours is 0.1215 well below 0.3 our data perfectly fit to predict new forms of data. Moving onto the training set(see image 25 graph on the right) we will need to look at some key areas. Firstly RMSE which we can see is 0.44998, R^2 is 0.9307, Adj R^2 is 0.9301, our goodness of fit which is at an f value of 1437.86 and a P value of <.0001 is good, and as seen before our residuals seem to fit with our assumptions and manages to satisfy the assumptions. In terms of our uber model of predicted prices this is ideal that our data will only be off on average by 0.44998 and can be used to explain 93.01% of the variation in our data. Our goodness of fit being high and significant meaning our variables are significant and at least one is. We can begin comparing the two tables using the RMSE, R^2, adj R^2. Overall, our train data performance seems to be better, having a better R^2, adj R^2 and RMSE. The difference between the two is not that drastic. This strongly suggests the model is not overfitting and will be able to generalize our uber data when new scenarios arise.

Finally with this model validated and cleaned the process can move onto the last step that being creating the prediction or rather the whole reason we made this model. I went ahead and made a single new scenario filling in the X variables our final model uses only and creating a prediction based on this. Again, noting that our Y variable currently is square rooted and these values to be real dollars would need to square which for this example I will do to demonstrate the model better. This gets us the following image (see image 26) which includes our predicted value and prediction interval. Our predicted value in the first row is 8.7320 with a prediction interval of (7.8328, 9.6313). Our model predicts the cost of our uber based off our input will result in a cost of $75.69. The predicted interval tells us the range the price should fall in ranges from $60.84 – $92.16. This is very good because we have a model that can predict the trip price and be off by relatively small cents difference instead of dollars, This uber model being able to predict prices so close suggests the model provides reasonably accurate estimates on how much an uber price would cost based on a certain number of key variables. We can provide a reasonable range of likely values for individual trips that can be accurate as seen in the column below the first being $36.2 and the prediction being $38 Perhaps with more data our model can continue to improve and increase the closeness of our real values while diminishing the error deviation. Overall, this is a strong starting model that can be used to predict Trip_Price with an efficiency at creating a predicted value and being close to the actual value.

Appendix

1.

Trip_Distance_km	Time_of_Day	Day_of_Week	Passenger_Count	Traffic_Conditic	Weather	Base_Fare	Per_Km_Rate	Per_Minute_Rate	Trip_Duration_M	Trip_Price
19.35	Morning	Weekday	3	Low	Clear	3.56	0.8	0.32	53.82	36.2624
47.59	Afternoon	Weekday	1	High	Clear		0.62	0.43	40.57	22.66190476
36.87	Evening	Weekend	1	High	Clear	2.7	1.21	0.15	37.27	52.9032
30.33	Evening	Weekday	4	Low	Rain	3.48	0.51	0.15	116.81	36.4698

2.

tod_afternoon	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	611	61.10	611	61.10
1	389	38.90	1000	100.00

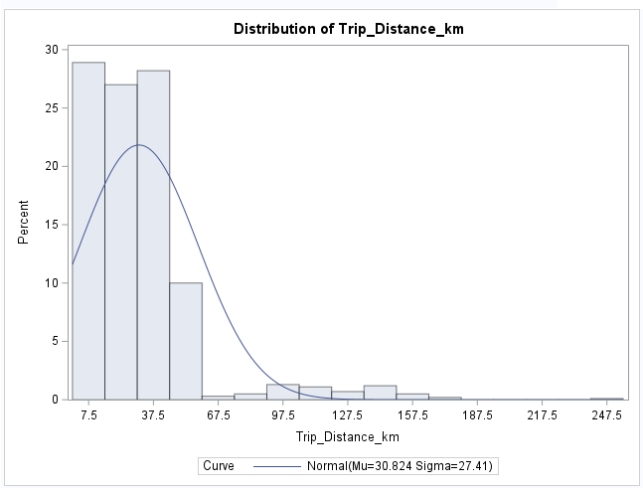
tod_evening	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	794	79.40	794	79.40
1	206	20.60	1000	100.00

tod_night	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	881	88.10	881	88.10
1	119	11.90	1000	100.00

dow_weekend	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	674	67.40	674	67.40
1	326	32.60	1000	100.00

traffic_medium	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	621	62.10	621	62.10
1	379	37.90	1000	100.00

3.



4.

Variable	Mean	Std Dev	Std Error	Lower 95% CL for Mean	Upper 95% CL for Mean	25th Pctl	50th Pctl	75th Pctl
Trip_Distance_km	30.8235270	27.4104611	0.8667949	29.1225795	32.5244745	13.0950000	26.9950000	40.3250000
Passenger_Count	2.4768421	1.1022491	0.0357617	2.4066610	2.5470232	1.0000000	2.0000000	3.0000000
Base_Fare	3.5029895	0.8701619	0.0282318	3.4475855	3.5583934	2.7300000	3.5200000	4.2600000
Per_Km_Rate	1.2333158	0.4298159	0.0139451	1.2059491	1.2606825	0.8600000	1.2200000	1.6100000
Per_Minute_Rate	0.2929158	0.1155918	0.0037503	0.2855560	0.3002756	0.1900000	0.2900000	0.3900000
Trip_Duration_Minutes	59.1406240	33.9292177	1.0729361	57.0351570	61.2460909	31.4700000	58.6400000	87.8000000
Trip_Price	54.6866574	40.6542526	1.2856003	52.1638705	57.2094442	31.4261000	48.0187000	67.5631000
tod_afternoon	0.3890000	0.4877673	0.0154246	0.3587318	0.4192682	0	0	1.0000000
tod_evening	0.2060000	0.4046328	0.0127956	0.1808906	0.2311094	0	0	0
tod_night	0.1190000	0.3239505	0.0102442	0.0988974	0.1391026	0	0	0
dow_weekend	0.3260000	0.4689818	0.0148305	0.2968975	0.3551025	0	0	1.0000000
traffic_medium	0.3790000	0.4853809	0.0153491	0.3488798	0.4091202	0	0	1.0000000
traffic_high	0.1990000	0.3994478	0.0126316	0.1742124	0.2237876	0	0	0
weather_rain	0.2310000	0.4216833	0.0133348	0.2048326	0.2571674	0	0	0
weather_snow	0.1010000	0.3014795	0.0095336	0.0822918	0.1197082	0	0	0

5.

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

	Trip_Price	Trip_Distance_km	Passenger_Count	Base_Fare	Per_Km_Rate	Per_Minute_Rate	Trip_Duration_Minutes	tod_afternoon	tod_evening	tod_night	dow_weekend	traffic_medium	traffic_high	weather_rain	weather_snow
Trip_Price	1.00000	0.61206	-0.01279	0.02721	0.25311	0.14315	0.20365	0.02031	0.00456	0.00200	-0.03366	-0.03265	0.09221	0.03604	0.01021
		<.0001	0.6937	0.4021	<.0001	<.0001	<.0001	0.5212	0.8854	0.9497	0.2876	0.3023	0.0035	0.2548	0.7470
	1000	1000	950	950	950	950	1000	1000	1000	1000	1000	1000	1000	1000	1000

6.

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	669644	47832	57.60	<.0001
Error	796	661002	830.40473		
Corrected Total	810	1330646			

Root MSE	28.81674	R-Square	0.5032
Dependent Mean	54.79028	Adj R-Sq	0.4945
Coeff Var	52.59463		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	-38.05560	6.75140	-5.64	<.0001
Trip_Distance_km	1	0.86467	0.03599	24.03	<.0001
Passenger_Count	1	-0.05854	0.92709	-0.06	0.9497
Base_Fare	1	1.47189	1.17580	1.25	0.2110
Per_Km_Rate	1	24.54942	2.37680	10.33	<.0001
Per_Minute_Rate	1	46.78940	8.75837	5.34	<.0001
Trip_Duration_Minutes	1	0.21525	0.03028	7.11	<.0001
tod_afternoon	1	3.23159	2.53451	1.28	0.2027
tod_evening	1	1.97144	2.95390	0.67	0.5047
tod_night	1	-0.54472	3.48456	-0.16	0.8758
dow_weekend	1	0.47019	2.17753	0.22	0.8291
traffic_medium	1	-0.51161	2.27376	-0.23	0.8220
traffic_high	1	8.02625	2.76056	2.91	0.0037
weather_rain	1	3.63711	2.43787	1.49	0.1361
weather_snow	1	2.91289	3.36857	0.86	0.3875

7.

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	-38.05560	6.75140	-5.64	<.0001	.	0
Trip_Distance_km	1	0.86467	0.03599	24.03	<.0001	0.98581	1.01439
Passenger_Count	1	-0.05854	0.92709	-0.06	0.9497	0.98311	1.01718
Base_Fare	1	1.47189	1.17580	1.25	0.2110	0.98925	1.01087
Per_Km_Rate	1	24.54942	2.37680	10.33	<.0001	0.98868	1.01145
Per_Minute_Rate	1	46.78940	8.75837	5.34	<.0001	0.97997	1.02044
Trip_Duration_Minutes	1	0.21525	0.03028	7.11	<.0001	0.98087	1.01950
tod_afternoon	1	3.23159	2.53451	1.28	0.2027	0.67179	1.48856
tod_evening	1	1.97144	2.95390	0.67	0.5047	0.71137	1.40573
tod_night	1	-0.54472	3.48456	-0.16	0.8758	0.76695	1.30386
dow_weekend	1	0.47019	2.17753	0.22	0.8291	0.98162	1.01872
traffic_medium	1	-0.51161	2.27376	-0.23	0.8220	0.83669	1.19518
traffic_high	1	8.02625	2.76056	2.91	0.0037	0.83285	1.20070
weather_rain	1	3.63711	2.43787	1.49	0.1361	0.95005	1.05258
weather_snow	1	2.91289	3.36857	0.86	0.3875	0.95188	1.05055

8.

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	669644	47832	57.60	<.0001
Error	796	661002	830.40473		
Corrected Total	810	1330646			

Root MSE	28.81674	R-Square	0.5032
Dependent Mean	54.79028	Adj R-Sq	0.4945
Coeff Var	52.59463		

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	398143	28439	304.73	<.0001
Error	728	67941	93.32591		
Corrected Total	742	466084			

Root MSE	9.66053	R-Square	0.8542
Dependent Mean	50.56461	Adj R-Sq	0.8514
Coeff Var	19.10533		

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	366473	26177	74.36	<.0001
Error	766	269642	352.01348		
Corrected Total	780	636116			

Root MSE	18.76202	R-Square	0.5761
Dependent Mean	50.72170	Adj R-Sq	0.5684
Coeff Var	36.99013		

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	396336	28310	407.18	<.0001
Error	715	49711	69.52628		
Corrected Total	729	446047			

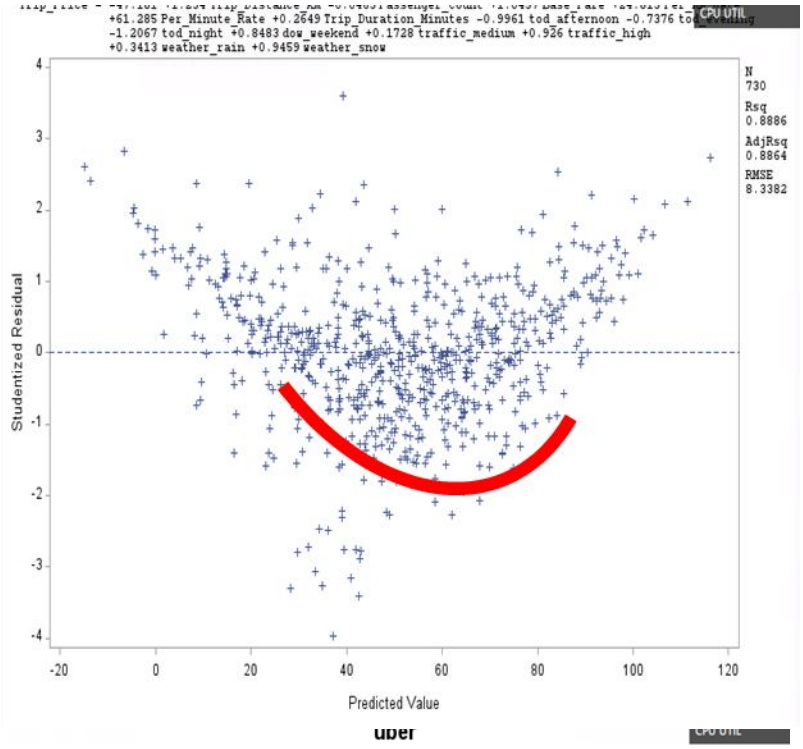
Root MSE	8.33824	R-Square	0.8886
Dependent Mean	50.85964	Adj R-Sq	0.8864
Coeff Var	16.39462		

Analysis of Variance

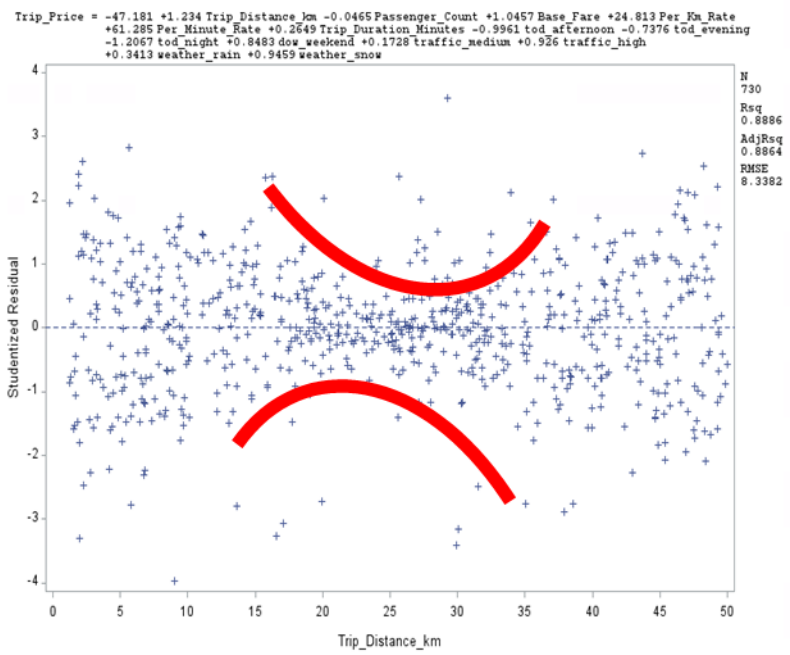
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	366289	26164	169.57	<.0001
Error	745	114948	154.29313		
Corrected Total	759	481237			

Root MSE	12.42148	R-Square	0.7611
Dependent Mean	50.02186	Adj R-Sq	0.7567
Coeff Var	24.83210		

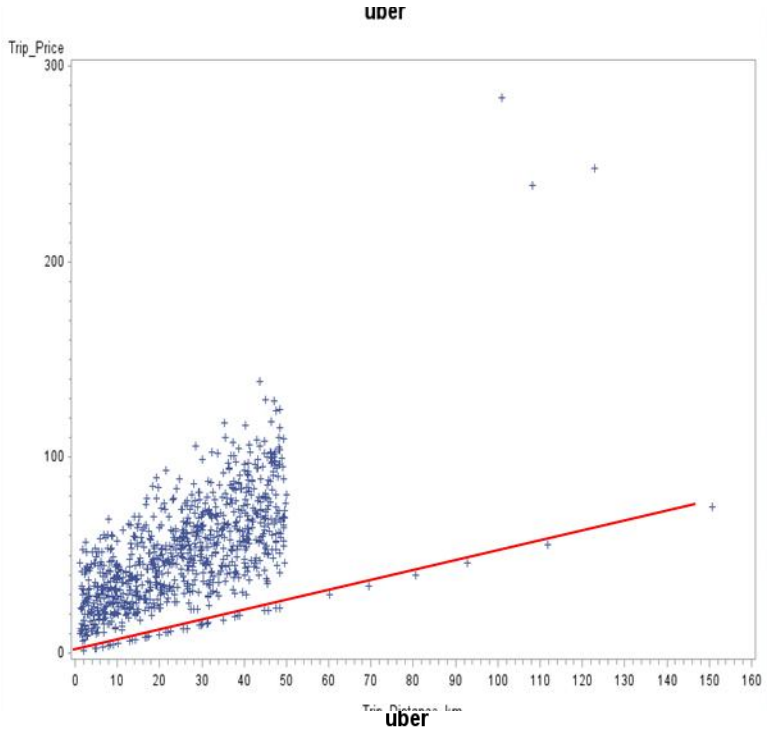
9.


+61.285 Per_Minute_Rate +0.2649 Trip_Duration_Minutes -0.9961 tod_afternoon -0.7376 tod
-1.2067 tod_night +0.8483 dow_weekend +0.1728 traffic_medium +0.926 traffic_high
+0.3413 weather_rain +0.9459 weather_snow
N
730
Rsq
0.8886
AdjRsq
0.8864
RMSE
8.3382
Studentized Residual
Predicted Value
uber

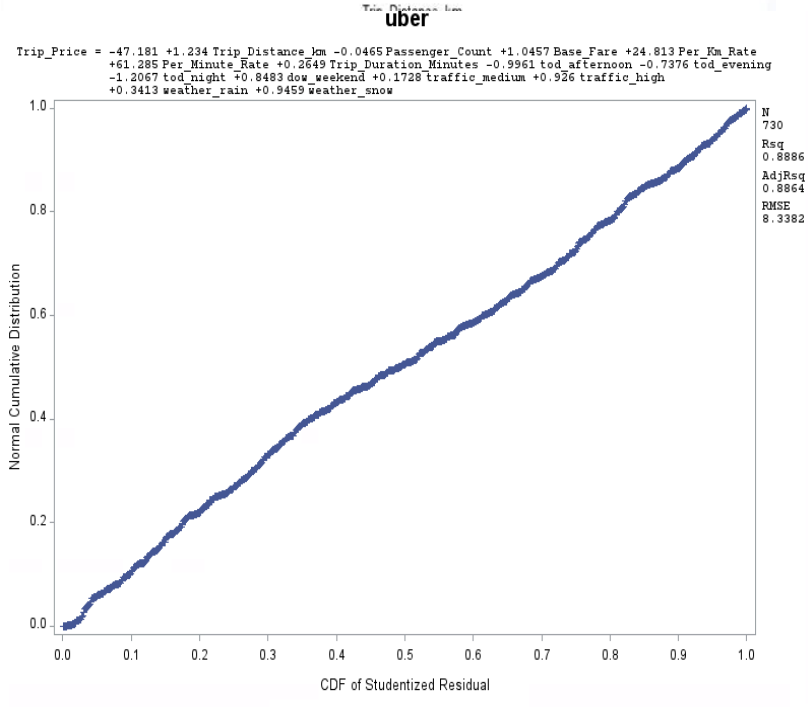
10.

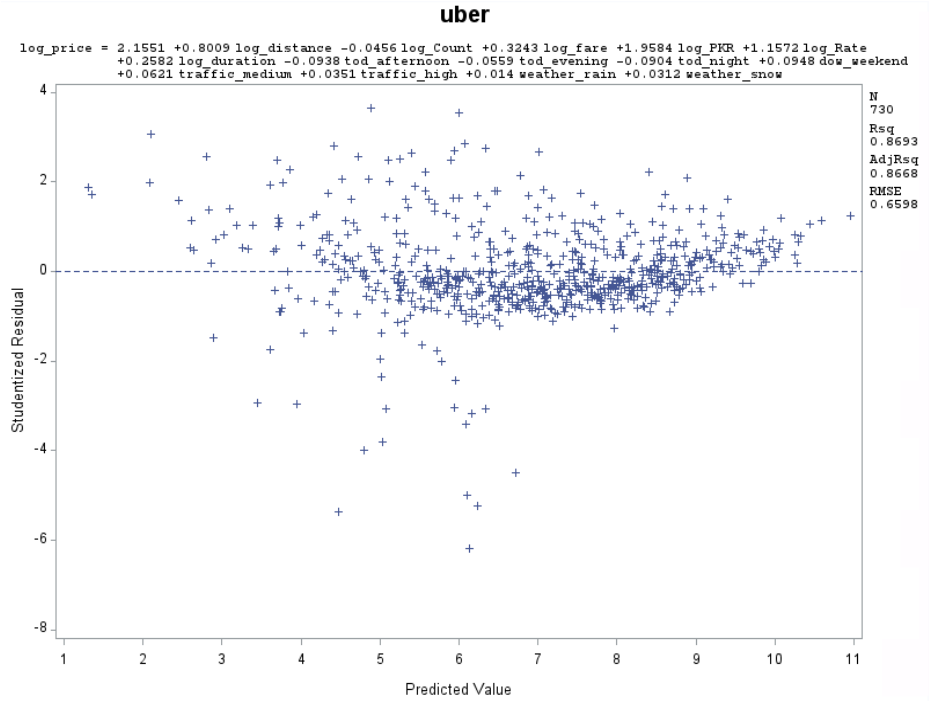

Trip_Price = -47.181 +1.234 Trip_Distance_km -0.0465 Passenger_Count +1.0457 Base_Fare +24.813 Per_Km_Rate
+61.285 Per_Minute_Rate +0.2649 Trip_Duration_Minutes -0.9961 tod_afternoon -0.7376 tod_evening
-1.2067 tod_night +0.8483 dow_weekend +0.1728 traffic_medium +0.926 traffic_high
+0.3413 weather_rain +0.9459 weather_snow
N
730
Rsq
0.8886
AdjRsq
0.8864
RMSE
8.3382
Studentized Residual
Trip_Distance_km

11.

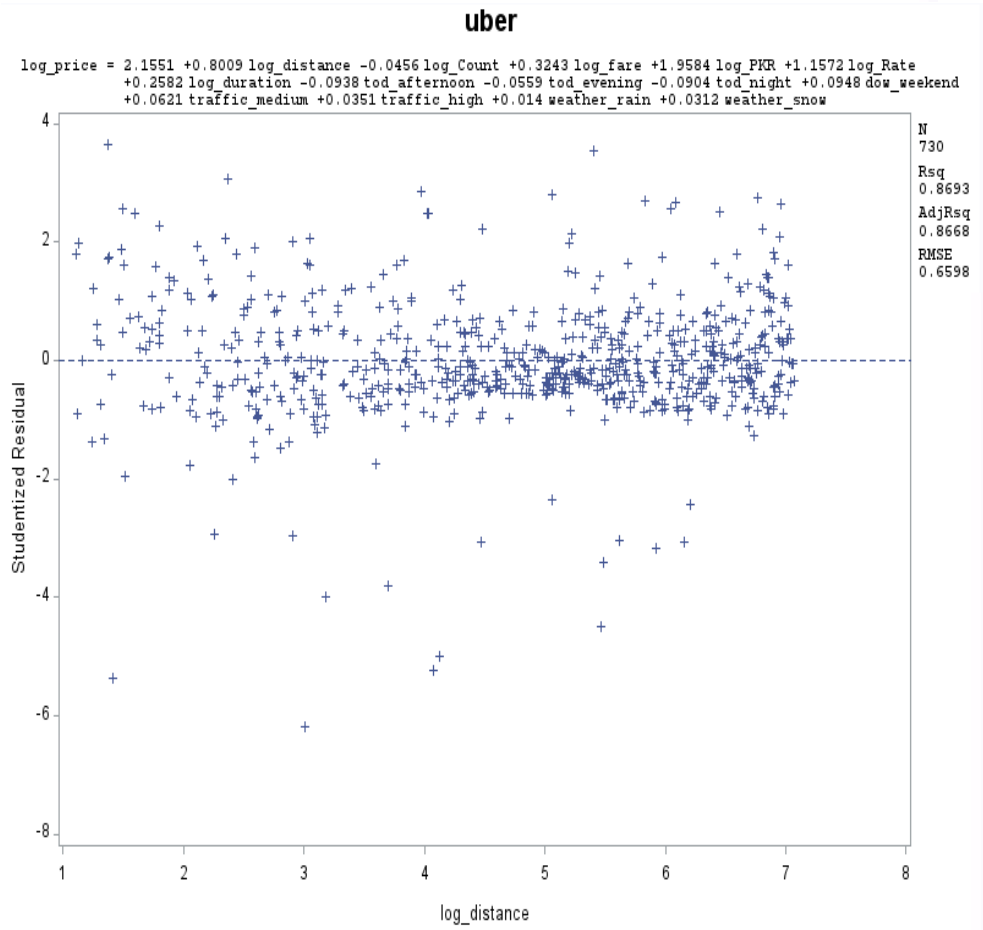

uber
Trip_Price
300
200
100
0
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160

12.

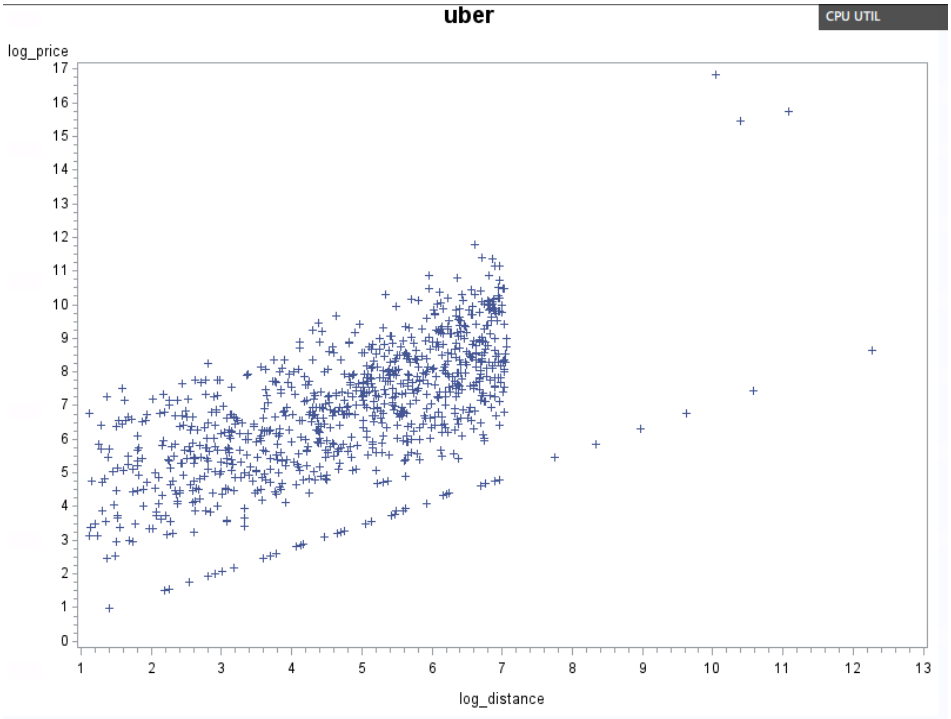

uber
Trip_Price = -47.181 +1.234 Trip_Distance_km -0.0465 Passenger_Count +1.0457 Base_Fare +24.813 Per_Km_Rate
+61.285 Per_Minute_Rate +0.2649 Trip_Duration_Minutes -0.9961 tod_afternoon -0.7376 tod_evening
-1.2067 tod_night +0.8483 dow_weekend +0.1728 traffic_medium +0.926 traffic_high
+0.3413 weather_rain +0.9459 weather_snow
N
730
Rsq
0.8886
AdjRsq
0.8864
RMSE
8.3382
Normal Cumulative Distribution
1.0
0.8
0.6
0.4
0.2
0.0
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
CDF of Studentized Residual


uber
log_price = 2.1551 +0.8009 log_distance -0.0456 log_Count +0.3243 log_fare +1.9584 log_PKR +1.1572 log_Rate
+0.2582 log_duration -0.0938 tod_afternoon -0.0559 tod_evening -0.0904 tod_night +0.0948 dow_weekend
+0.0621 traffic_medium +0.0351 traffic_high +0.014 weather_rain +0.0312 weather_snow
N
730
Rsq
0.8693
AdjRsq
0.8668
RMSE
0.6598
Studentized Residual
Predicted Value

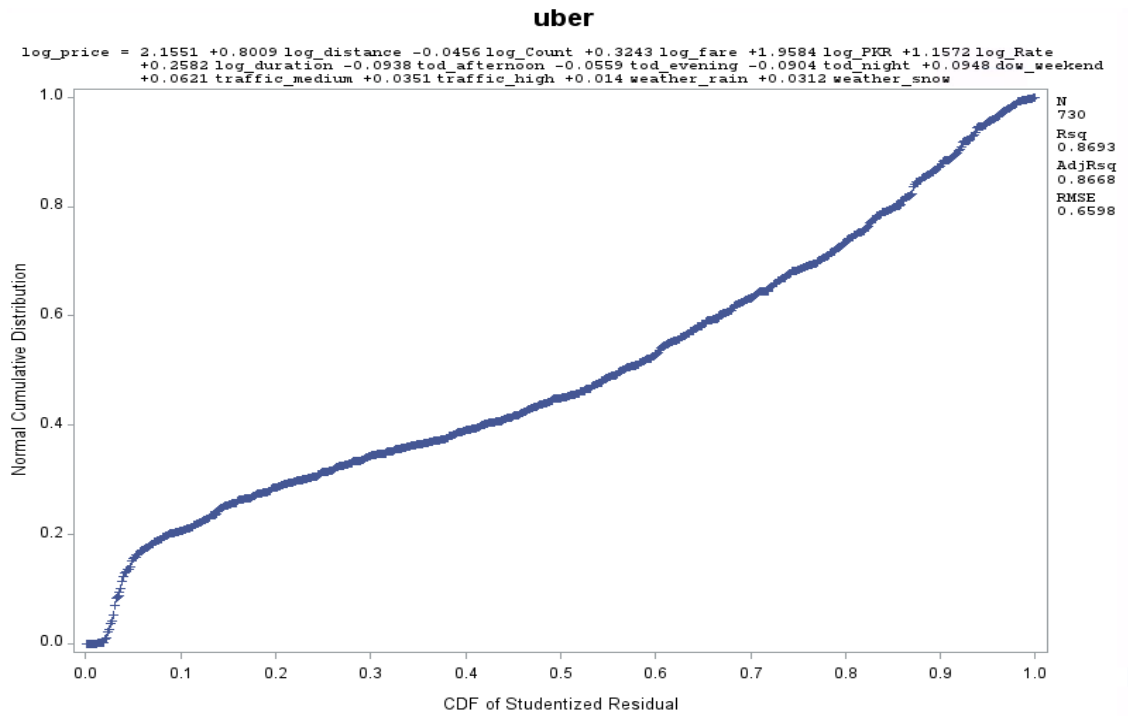
13.


uber
log_price = 2.1551 +0.8009 log_distance -0.0456 log_Count +0.3243 log_fare +1.9584 log_PKR +1.1572 log_Rate
+0.2582 log_duration -0.0938 tod_afternoon -0.0559 tod_evening -0.0904 tod_night +0.0948 dow_weekend
+0.0621 traffic_medium +0.0351 traffic_high +0.014 weather_rain +0.0312 weather_snow
N
730
Rsq
0.8693
AdjRsq
0.8668
RMSE
0.6598
Studentized Residual
log_distance

14.


uber
CPU UTIL
log_price
log_distance

15.


uber
log_price = 2.1551 +0.8009 log_distance -0.0456 log_Count +0.3243 log_fare +1.9584 log_PKR +1.1572 log_Rate
+0.2582 log_duration -0.0938 tod_afternoon -0.0559 tod_evening -0.0904 tod_night +0.0948 dow_weekend
+0.0621 traffic_medium +0.0351 traffic_high +0.014 weather_rain +0.0312 weather_snow
N
730
Rsq
0.8693
AdjRsq
0.8668
RMSE
0.6598
Normal Cumulative Distribution
CDF of Studentized Residual

16.

17.

Forward Selection: Step 9

Variable log_Count Entered: R-Square = 0.8797 and C(p) = 5.7614

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	1500.82249	166.75805	433.21	<.0001
Error	533	205.17027	0.38493		
Corrected Total	542	1705.99276			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	2.30886	0.19675	53.00807	137.71	<.0001
log_distance	0.78998	0.01695	836.16385	2172.22	<.0001
log_Count	-0.04371	0.05335	0.25834	0.67	0.4130
log_fare	0.24784	0.10412	2.18112	5.67	0.0176
log_PKR	2.01049	0.07048	313.19103	813.62	<.0001
log_Rate	1.19671	0.05991	153.60506	399.04	<.0001
log_duration	0.26091	0.01039	242.81897	630.81	<.0001
tod_night	-0.08800	0.07908	0.47661	1.24	0.2663
dow_weekend	0.07619	0.05679	0.69285	1.80	0.1803
traffic_medium	0.04665	0.05456	0.28143	0.73	0.3929

Backward Elimination: Step 9

Variable dow_weekend Removed: R-Square = 0.8787 and C(p) = 2.1222

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1499.13040	299.82608	778.33	<.0001
Error	537	206.86236	0.38522		
Corrected Total	542	1705.99276			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	2.33582	0.18676	60.25779	156.42	<.0001
log_distance	0.78917	0.01691	838.51942	2176.74	<.0001
log_fare	0.23662	0.10391	1.99771	5.19	0.0232
log_PKR	1.99770	0.07008	313.03829	812.63	<.0001
log_Rate	1.19595	0.05984	153.85653	399.40	<.0001
log_duration	0.25935	0.01036	241.41937	626.71	<.0001

18.

Number of Observations Read	917
Number of Observations Used	573
Number of Observations with Missing Values	344

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1536.84596	307.36919	531.57	<.0001
Error	567	327.85813	0.57823		
Corrected Total	572	1864.70409			

Root MSE	0.76042	R-Square	0.8242
Dependent Mean	6.98449	Adj R-Sq	0.8226
Coeff Var	10.88721		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	2.22538	0.22097	10.07	<.0001
log_distance	1	0.76965	0.01955	39.37	<.0001
log_fare	1	0.27760	0.12423	2.23	0.0258
log_PKR	1	1.88472	0.08397	22.44	<.0001
log_Rate	1	1.08919	0.07161	15.21	<.0001
log_duration	1	0.26125	0.01231	21.22	<.0001

19.

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	2.22538	0.22097	10.07	<.0001	.	0
log_distance	1	0.76965	0.01955	39.37	<.0001	0.99500	1.00502
log_fare	1	0.27760	0.12423	2.23	0.0258	0.99807	1.00193
log_PKR	1	1.88472	0.08397	22.44	<.0001	0.99387	1.00616
log_Rate	1	1.08919	0.07161	15.21	<.0001	0.99287	1.00718
log_duration	1	0.26125	0.01231	21.22	<.0001	0.99131	1.00877

20.

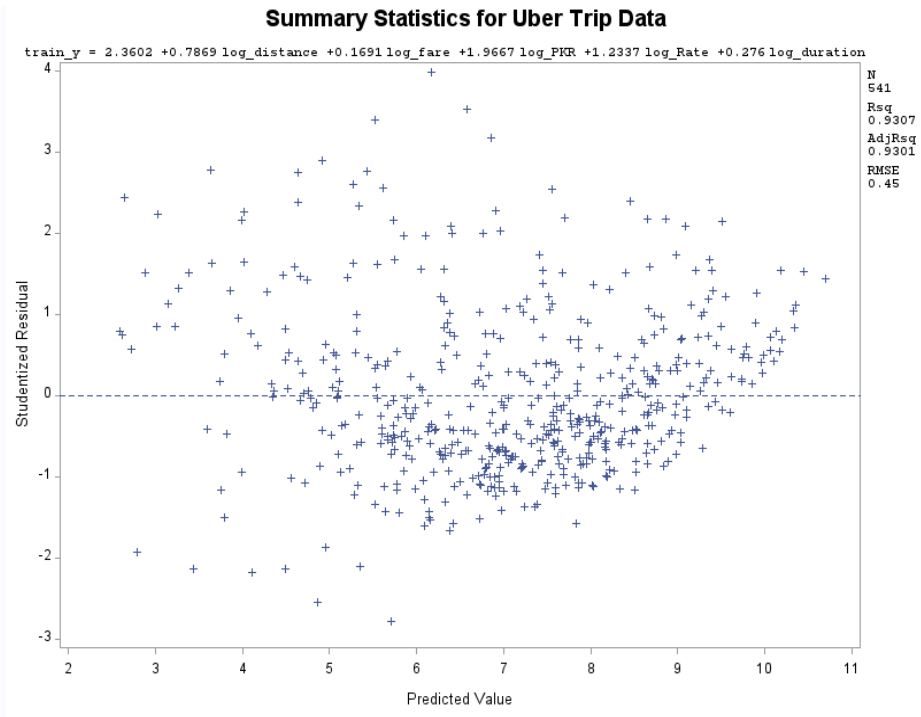
Root MSE	0.53792	R-Square	0.9049
Dependent Mean	7.01648	Adj R-Sq	0.9040
Coeff Var	7.66646		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	2.44789	0.16153	15.15	<.0001
log_distance	1	0.78277	0.01441	54.32	<.0001
log_fare	1	0.19433	0.08952	2.17	0.0304
log_PKR	1	2.00376	0.06033	33.21	<.0001
log_Rate	1	1.20883	0.05182	23.33	<.0001
log_duration	1	0.25852	0.00910	28.41	<.0001

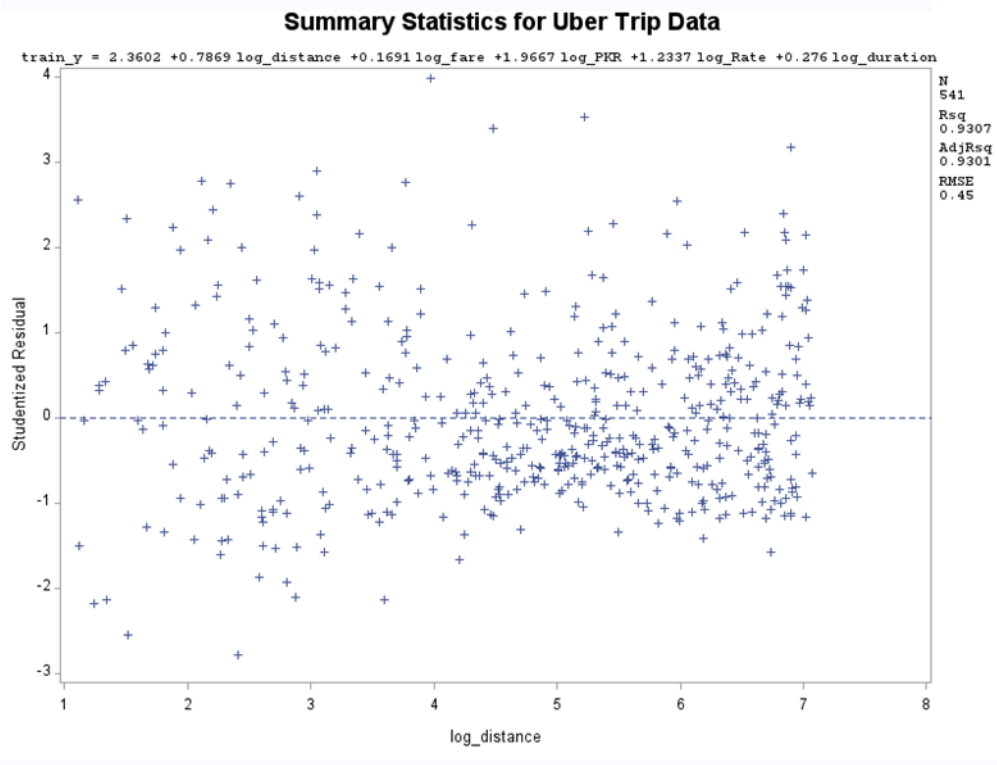
Root MSE	0.44998	R-Square	0.9307
Dependent Mean	7.06377	Adj R-Sq	0.9301
Coeff Var	6.37020		

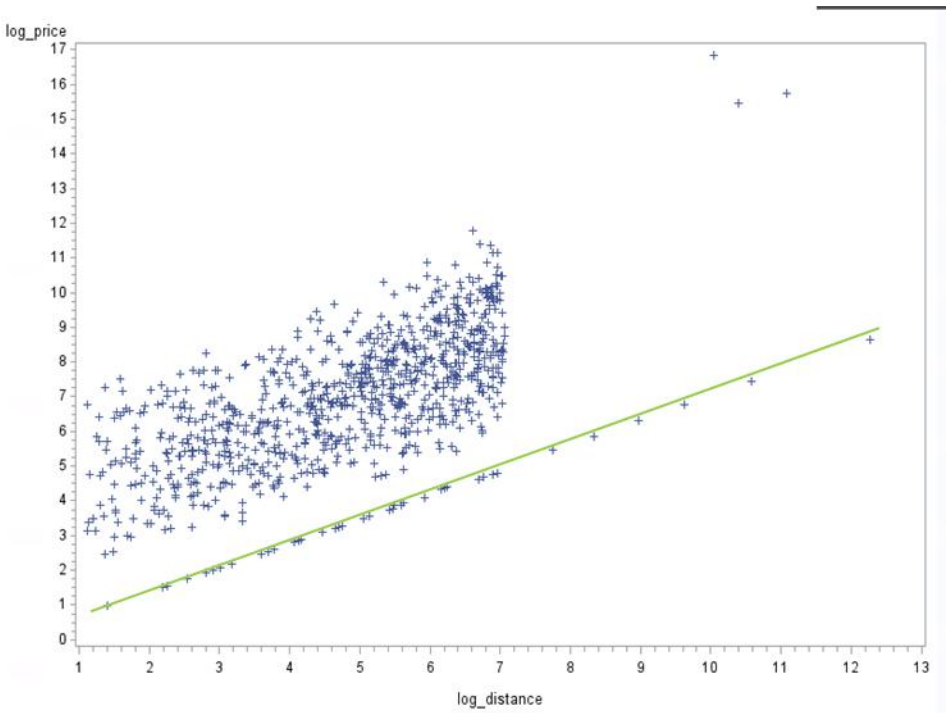
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	2.36016	0.13689	17.24	<.0001
log_distance	1	0.78693	0.01228	64.09	<.0001
log_fare	1	0.16910	0.07575	2.23	0.0260
log_PKR	1	1.96670	0.05152	38.17	<.0001
log_Rate	1	1.23366	0.04438	27.80	<.0001
log_duration	1	0.27598	0.00804	34.32	<.0001

21.

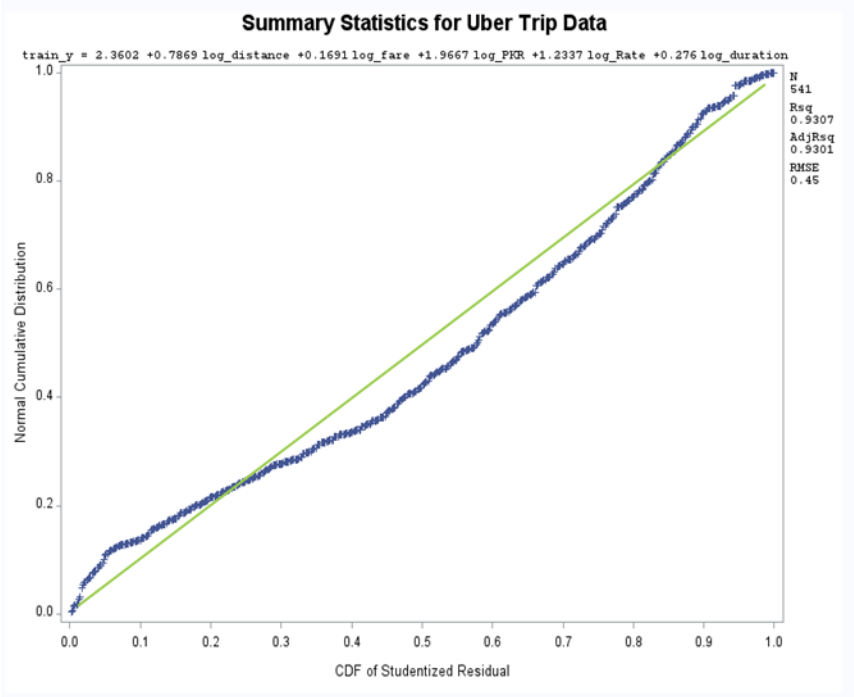

Summary Statistics for Uber Trip Data
train_y = 2.3602 +0.7869 log_distance +0.1691 log_fare +1.9667 log_PKR +1.2337 log_Rate +0.276 log_duration
N
541
Rsq
0.9307
AdjRsq
0.9301
RMSE
0.45
Studentized Residual
Predicted Value

22.


Summary Statistics for Uber Trip Data
train_y = 2.3602 +0.7869 log_distance +0.1691 log_fare +1.9667 log_PKR +1.2337 log_Rate +0.276 log_duration
N
541
Rsq
0.9307
AdjRsq
0.9301
RMSE
0.45
Studentized Residual
log_distance


log_price
log_distance

23.


Summary Statistics for Uber Trip Data
train_y = 2.3602 +0.7869 log_distance +0.1691 log_fare +1.9667 log_PKR +1.2337 log_Rate +0.276 log_duration
N
541
Rsq
0.9307
AdjRsq
0.9301
RMSE
0.45
Normal Cumulative Distribution
CDF of Studentized Residual

24.

25.

Summary Statistics for Uber Trip Data

Obs	_TYPE_	_FREQ_	rmse	mae
1	0	229	0.82227	0.51357

Summary Statistics for Uber Trip Data

The CORR Procedure

2 Variables:	log_price yhat

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
log_price	229	6.73860	1.84997	1543	0.97101	11.77465	
yhat	197	6.73787	1.79214	1327	1.19423	13.15801	Predicted Value of train_y

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

	log_price	yhat
log_price	1.00000 229	0.89965 <.0001 197
yhat Predicted Value of train_y	0.89965 <.0001 197	1.00000 197

Number of Observations Read	885
Number of Observations Used	541
Number of Observations with Missing Values	344

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1455.68224	291.13645	1437.86	<.0001
Error	535	108.32621	0.20248		
Corrected Total	540	1564.00846			

Root MSE	0.44998	R-Square	0.9307
Dependent Mean	7.06377	Adj R-Sq	0.9301
Coeff Var	6.37020		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	2.36016	0.13689	17.24	<.0001
log_distance	1	0.78693	0.01228	64.09	<.0001
log_fare	1	0.16910	0.07575	2.23	0.0260
log_PKR	1	1.96670	0.05152	38.17	<.0001
log_Rate	1	1.23366	0.04438	27.80	<.0001
log_duration	1	0.27598	0.00804	34.32	<.0001

26.

log_distance	log_fare	log_PKR	log_Rate	log_duration	Selected	Trip_Distance_km	Time_of_Day	Day_of_Week	log_price	log_Count	train_y	phat	lcl_pred	ucl_pred
3.5000	1.20000	0.30000	0.50000	8.0000	.	.			.	.		8.7320	7.8328	9.6313
4.3989	1.26976	-0.22314	-1.13943	7.3362	1	19.35	Morning	Weekday	6.0218	1.09861	6.0218	6.2166	5.3308	7.1024
6.8986	.	-0.47804	-0.84397	6.3695	1	47.59	Afternoon	Weekday	4.7605	0.00000	4.7605	.	.	.